

Chapter 53 | Population Ecology

General Biology I · Unit 8: Ecology · First-year undergraduate

53.1 Describing a Population

A population is a group of individuals of the same species living in the same area and capable of interbreeding. Population ecology asks a quantitative question about such a group: how does its size change over time, and what limits it?

Population size is written as N . The rate of change of that size is written dN/dt , read as the change in N with respect to time. When dN/dt is positive the population is growing. When it is zero the population is stable.

53.2 Exponential Growth

A population in an environment with unlimited resources grows at a rate proportional to its own size. Each individual contributes to reproduction, so more individuals means faster growth. This produces the exponential growth model.

$$dN/dt = rN$$

The constant r is the intrinsic rate of increase, sometimes called the per capita growth rate. It is the difference between the per individual birth rate and the per individual death rate. A larger r produces a steeper curve.

Exponential growth is real but temporary. Bacteria in fresh medium, or a species newly introduced to an island, will follow this curve for a time. No population follows it indefinitely, because no environment has unlimited resources.

53.3 Carrying Capacity

Every environment can support only a finite number of individuals. That maximum sustainable population size is the carrying capacity, written K . It is set by whatever resource runs out first, which may be food, water, nesting sites, or space itself.

As N approaches K , competition intensifies. Birth rates fall, death rates rise, and growth slows. When N reaches K , births and deaths balance and dN/dt falls to zero.

53.4 The Logistic Growth Model

The logistic model modifies exponential growth by adding a term that shuts growth down as the population fills the available space. It is the central equation of this chapter.

$$dN/dt = rN (1 - N/K)$$

The bracketed term is the key. When N is very small compared with K , the fraction N/K is close to zero, the bracket is close to one, and the equation behaves almost exactly like exponential growth. When N approaches K , the fraction approaches one, the bracket approaches zero, and growth stops.

The result is an S shaped curve, often called a sigmoid curve.

The steepest growth occurs at $N = K/2$. This is the inflection point of the logistic curve, and it is where the population is adding individuals fastest in absolute terms.

53.5 Reading the Two Parameters

The two parameters control different features of the curve, and students routinely confuse them.

The value of r controls how quickly the population reaches its ceiling. It changes the steepness of the rise but not the final level. A population with a high r climbs sharply and flattens early. A population with a low r climbs gently and may still be rising after many generations.

The value of K controls where the curve flattens. It sets the ceiling itself. Changing K raises or lowers the plateau without changing how fast the population approaches it.

A useful consequence follows. A population can reach a given size at a given time through very different combinations of these two parameters. A fast growing population with a low ceiling and a slower growing population with a high ceiling may pass through the same point.

53.6 Worked Example

A population of deer is introduced to a reserve. The starting population is 20 individuals. Field studies estimate r at 0.30 per year and K at 500 individuals.

At $N = 20$: $dN/dt = 0.30 \times 20 \times (1 - 20/500) = 5.76$ deer per year.

At $N = 250$: $dN/dt = 0.30 \times 250 \times (1 - 250/500) = 37.5$ deer per year.

At $N = 480$: $dN/dt = 0.30 \times 480 \times (1 - 480/500) = 5.76$ deer per year.

Notice that the growth rate is the same at 20 individuals and at 480 individuals, for entirely different reasons. Early on there are too few individuals reproducing. Late on there are too few resources remaining. The population grows fastest in the middle.

53.7 Density Dependence

Factors that limit a population more strongly as density rises are called density dependent.

Competition, predation, disease transmission, and waste accumulation all fall into this category, and all are captured by the bracketed term in the logistic equation. Density independent factors, such as a sudden freeze or a flood, reduce population size regardless of how crowded the population was.

Practice

1. A population has $r = 0.5$ and $K = 1000$. Calculate dN/dt at $N = 100$, $N = 500$, and $N = 900$.
2. Two populations reach 400 individuals after 25 years. One has $r = 0.2$, the other $r = 0.6$. What must be true of their carrying capacities? Explain.
3. Sketch the logistic curve for $r = 0.4$ and $K = 800$ over 40 years starting from $N = 10$. Mark the inflection point.

4. Explain why raising r does not raise the eventual population size.